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Editor-in-Chief
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Dear Editor,

We are pleased to submit our original research article entitled "**A Deployment-Safety-Oriented Framework for ECG Model Calibration Transferability: A 60-Experiment Boundary Validation Across Three Public Corpora**" for your consideration as a **Full Length Article** in *Computer Methods and Programs in Biomedicine*. This manuscript has not been published elsewhere and is not under consideration by any other journal.

**Summary of contributions.** The manuscript contributes three reusable method components — a three-component (slope, intercept, prevalence) Shapley decomposition of the transfer benefit, a two-layer deployment-safety gate S1/S2, and a pre-registered applicability-boundary protocol (amendments A1–A4) — validated end-to-end by the temperature-scaling study summarized below. Temperature scaling (TS) is the default post-hoc recalibration for deep classifiers, yet whether its in-distribution (ID) benefit transfers to out-of-distribution (OOD) target corpora remains the central open question for cross-institution ECG deployment. We pre-registered and executed a multi-seed calibration boundary study comprising **60 experiments** (6 directed transfer pairs $\times$ 2 architectures $\times$ 5 seeds, fully balanced) on PTB-XL, Chapman-Shaoxing, and CPSC2018, with a patient-level cluster paired bootstrap (the operating CI is BCa at $B = 10{,}000$, fully re-run on the entire 60-experiment grid) and eight recalibration methods. TS yields a positive OOD calibration benefit $\Delta\mathrm{ECE}_{\mathrm{OOD}} = \mathrm{ECE}_{raw}^{OOD} - \mathrm{ECE}_{TS}^{OOD}$ in **45/60 = 75.0%** of seed experiments (InceptionTime 24/30, ResNet1D 21/30), while the remaining **15 counter-examples are transparently disclosed** and mechanistically explained case by case (seven have OOD accuracy $\leq 0.50$, seven have an ID–OOD accuracy drop $\geq 0.18$, and six satisfy both; the remainder are seed-level temperature mis-transfers, one of which is over-correction on an already well-calibrated backbone). The paper delivers **three contributions**:

1. **A reusable deployment-safety evaluation framework:** a three-component Shapley decomposition of the transfer benefit (**C3**; 27/27 cells at the pre-registered $n = 20{,}000$ readout), a two-layer deployment-safety gate S1/S2 (**C5**; S1: positive calibration benefit, S2: OOD accuracy at or above the target majority-class baseline), and a pre-registered applicability-boundary protocol (amendments A1–A4) that fixes what may be claimed before unblinding.
2. **A 60-experiment boundary characterization as the validation carrier** (**C1**, **C2**, **C4**): after a disclosed validation-leak repair (20/60 cells retrained), TS yields a positive OOD benefit in 45/60 seed experiments (75.0%; InceptionTime 24/30, ResNet1D 21/30),

with fifteen fully disclosed counter-examples; the registered ID-boundary criterion is not met ($30/60 = 50.0\% < 80\%$) and its branch fires (OOD benefit 2.7× the ID benefit); TS is the most consistent of the eight delivered recalibrators, EM prior harmful on average ($-0.0040$ delivered, $-0.0378$ per-branch) and Matrix scaling fragile (24/60); the gate passes 6/12 strata at the stratum-mean level (5/12 per-seed majority); the strict post-calibration deployment bar is met on 0/60 cells — by global TS and by the stronger per-sample ATS (+0.0190) alike.

3. **A measured, verifiable deployment path:** buying 50 labeled target records and recalibrating locally yields +0.150, an order of magnitude above the zero-shot benefit; no label-free gate is claimed (the registered failure branch); released code, patient-level splits, hash-locked tables, and a 12-check verification gate make every reported number recomputable from public artifacts.

**Why this journal.** Cross-corpus ECG transfer calibration is a critical bottleneck for clinical deployment of deep ECG models across institutions with different acquisition protocols, lead sets, and label vocabularies. Our contribution is evaluated end-to-end on the raw 12-lead ECG signal: deep classifiers trained on one corpus are pushed through an eval-time shift ladder (sampling rate, lead availability, noise, gain) and a six-direction cross-corpus transfer grid, and the contribution is a boundary-condition characterization of when post-hoc recalibration pays off at deployment—combined with transparent counter-example reporting, a discrimination-aware deployment gate, and a pre-registered failure branch. We believe this falls squarely within the journal's scope on biomedical signal processing, reliability and calibration of clinical biosignal models, and reproducible evaluation methodology.

**Transparency statement.** All hypotheses in this manuscript are tagged *exploratory + robustness validation*, not confirmatory, consistent with the open-science best practices of Nosek et al. (2018). The fifteen counter-examples (25% of the experiment grid) are disclosed in full in the manuscript: the 60-seed results table reports every per-seed $\Delta$ECE with its 95% CI lower bound and every counter-example's OOD accuracy, and each counter-example is mechanistically explained case by case. Four pre-registration amendments are recorded with reason and date: A1 re-locates the primary endpoint (protocol v2.0 $\rightarrow$ v2.1-A1), A2 reduces the confirmatory family from 156 to 12 hypotheses, re-designating the L2 shift levels as exploratory dose-response, A3 corrects the identity of the third corpus (the registered "CPSC2018+2019" is in fact *CPSC2018 only* – the CPSC Database plus the CPSC-Extra Database) and discloses that CPSC2018's official nine-class taxonomy contains no MI, so the MI class of our four-class subspace is populated entirely by CPSC-Extra SNOMED annotations, and A4 (2026-10-11) re-operationalizes the protocol's registered mapping-sensitivity gate on the delivered primary endpoint: the registered wording conditioned the gate on the secondary endpoint $G$, which its own Limitations acknowledge as undelivered on real data, so the gate now reads cross-pair $\Delta\text{ECE}_{\text{OOD}}$ ordering (Kendall $\tau \geq 0.8$) with the confirmatory conclusions unchanged, defined *before* any mapping-variant result exists. A3 changes no recorded value (naming, attribution and disclosure); A4 changes no recorded value (gate operationalization only; the variant battery remains a queued deliverable). The protocol and the amendments are archived as an immutable record with per-file SHA-256 checksums at `https://doi.org/10.17605/OSF.IO/53H64` (amendments A3 and A4 are deposited in the OSF project deposit – project 53h64 Files, 2026-10-11, per-file SHA-256 verified; the frozen registration WJZ3Y remains intentionally at v1, the amendment set lives in the project deposit). A five-item honesty declaration (adversarial audit product) is included in the manuscript.

**Data and code availability.** The full pipeline code, the patient-level split indices, and the per-experiment result artifacts are already publicly available at `https://github.com/Toni-cs/ecg-calibration-transportability`, with trained model checkpoints for the 60-experiment grid provided as release assets of the same repository. The repository will be archived on Zenodo with a DOI upon acceptance. All three ECG corpora are public (PhysioNet/Kaggle).

This is a five-author submission; the authors share responsibility for all aspects of the work (see the CRediT statement in the manuscript). Highlights, the CRediT contribution statement, and a data statement are provided with the manuscript. We look forward to your feedback.

Sincerely,

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